GIOVANNI A. CAROSSO, PhD

Bioengineer | Biotech Operator | Platform Strategy

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PROFESSIONAL SUMMARY

Bioengineer and biotech operator building ML-enabled modality platforms for epigenetic medicine. Led R&D from inception to multiple positive proofs-of-concept *in vivo*, enabling two financing rounds and a Novo Nordisk strategic partnership. Architected platforms behind EPI-321 (clinical-stage FSHD), EPI-331 (DMD), and a benchmark-beating pre-clinical pipeline of single-dose editors across diverse genetic targets (cardiometabolic, muscular, longevity). Deep domain expertise in epigenetic editing, protein LMs/engineering, predictive modeling, functional genomics, mRNA-LNP programs, target selection, technical strategy, and investor/pharma-partner diligence.

TECHNICAL SKILLS

Machine learning: self-supervised protein LMs (ESM-2 fine-tuning; pLM-guided MCMC over sequence space), deep generative design (Boltzgen, structure-conditioned), structure prediction (AlphaFold2-Multimer, Boltz-2, ESMFold), supervised sequence–function models, few-shot active learning, chromatin state classifiers (ChromHMM), Python, R.

Platform architecture/discovery: high-throughput CRISPR activation/inhibition, gene editing, genome-scale perturbation, proteomics & transcriptomics, multi-axis hit scoring, offline metrics & evals, NGS (RNA-seq, Hi-C, Perturb-seq), tech/IP strategy.

Payload/drug development: protein/mRNA engineering, *in vivo* mRNA-LNP delivery, *in vivo* pharmacology, pre-clinical trial design.

EXPERIENCE

General Control
Jul 2024 – Jul 2026

Founding Head of Research

- Built R&D platform delivering a best-in-class, novel epigenetic editor IP portfolio; achieved *in vivo* proof-of-concept in 12 months and advanced 6/7 pipeline programs to positive pre-clinical readouts in under 24 months, by resolving a hierarchical mechanism of epigenetic memory that unlocked novel targets for one-shot mRNA-LNP dosing.
- Defined scientific strategy from pre-seed/concept and executed technical de-risk behind two capital raises and Novo Nordisk partnership; lead inventor on platform PCTs.
- Architected ML stack (protein LMs, chromatin-state classifiers) and deployed it to generate novel editors and *de novo* binders; screened 10,000-element libraries; recovered novel hits *in vitro* and *in vivo*.
- Scaled discovery throughput ~100× with a combinatorial peptide platform, isolating editors that outperform benchmarks across diverse genetic targets; advanced top trial candidates by predictive modeling.

- Delivered >90% simultaneous knockdown of up to three targets with a novel mRNA-LNP liver-targeted payload *in vivo*, establishing multiplexed one-shot gene control as a platform differentiator.

Epicrispr Biotechnologies
Jan 2021 – Jul 2024

Senior Scientist (prev. Scientist II, Scientist)

- Co-developed Epic's GEMS modality platform; engineered proprietary epigenetic editors including EPI-321, which has reported statistically significant gains in lean muscle volume, DUX4-suppression biomarkers, no serious adverse events in its first-in-human FSHD trial (NCT06907875). EPI-321 gained FDA IND clearance, Fast Track, Rare Pediatric Disease, and Orphan Drug designations, and drove $68M Series B followed by $90M Series C with crossover syndicate.
- Lead author on hypercompact activator discovery (64–98 residues; five-week efficacy via single-dose mRNA-LNP), cited in *Nature Reviews Drug Discovery* (2025). Led protein-engineering campaigns producing the first-described mitotically durable gene activators; ~10× benchmark duration *in vitro* and *in vivo* via LNP delivery.
- Established Applied Technologies team; raised pooled-screen hit rates by ~8×; generated the platform's structured ML training corpus behind few-shot learning-based protein engineering with evolutionary sampling. *NeurIPS* (2023).

UCSF Dept. of Neurological Surgery
Jan 2020 – Dec 2020

Postdoctoral Scholar

- Integrated Perturb-seq, PLAC-seq, ChIP-seq and Hi-C into a machine-learning classifier over genome-scale dual CRISPRi screens spanning 18,905 coding and 10,678 lncRNA loci, with the Weissman and Lim labs. *Cell Genomics* (2022).

EDUCATION

PhD, Human Genetics
2019

Johns Hopkins University School of Medicine

- Discovered molecular drivers in pediatric disorders of chromatin machinery (Hans Bjornsson Lab), novel roles for KMT2D in oxygen sensing during neurodevelopment. Dissertation, *JCI Insight* (2019).
- Secured NASA funding award for astronaut epigenetics in the ISS Twins Study Consortium (Andy Feinberg Lab); co-developed first protocols for in-flight epigenomic sampling aboard ISS and experiments for astronauts Scott and Mark Kelly.
- Led a US Dept. of Defense-funded neuroscience program in Bolivia (Clubes de Ciencia Bolivia); published on roles for scientists as non-state actors of international diplomacy. *Nature Human Behaviour* (2019), *Nature Communications* (2020).

BS, Biopsychology
2010

UC Santa Barbara

- Behavioral neurobiology; published on mechanisms of dopaminergic dysfunction in drug abuse. *Addiction Biology* (2012).
- Discovery of *POLR3A* mutations causing pediatric leukodystrophy (Children's National Medical Center). *AJHG* (2011).

COMMUNICATION

- Languages: Italian, French, Spanish, English.

PATENTS

US 2024/0216482 A1 — Systems and methods for regulating aberrant gene expressions. Epicrispr Biotechnologies, Inc. App. 18/542,396; priority 2021-06-17; published 2024-07-04; pending.

US 2024/0254659 A1 — Systems and methods for regulating target genes. Epicrispr Biotechnologies, Inc. App. 18/417,827; priority 2021-07-20; published 2024-08-01; pending.

WO 2023/172995 A1 — Systems and methods for genetic modulation to treat ocular diseases. Epicrispr Biotechnologies, Inc. App. PCT/US2023/064004; priority 2022-03-11; published 2023-09-14; pending.

US 2025/0388634 A1 — Engineered gene effectors, compositions, and methods of use thereof. Epicrispr Biotechnologies, Inc. App. 18/842,718; priority 2022-03-24; published 2025-12-25; pending.

US 2026/0085300 A1 — Systems and compositions for fusion polypeptides and methods of use thereof. Epicrispr Biotechnologies, Inc. App. 19/123,019; priority 2022-10-27; published 2026-03-26; pending.

WO 2024/238661 A2 — Systems and methods for regulating target genes. Epicrispr Biotechnologies, Inc. Priority 2023-05-17; published 2024-11-21; pending.

PCT/US26/24038 — Compositions and Methods for Epigenome Editing. Epi Labs, Inc. Filed 2026-04-16.

PCT/US26/24051 — Systems and Methods for Upregulation of Exerkines. Epi Labs, Inc. Filed 2026-04-16.

Additional US provisional and PCT filings pending publication.

PUBLICATIONS

1. **Carosso, G.A.**, Yeo, R.W., Gainous, T.B., Jawaid, Z., Yang, X., Cutillas, V., Qi, L.S., Daley, T.P., Hart, D. (2024). Discovery of hypercompact epigenetic modulators for persistent CRISPR-mediated gene activation. *bioRxiv*.
2. Wu, D., Poddar, A., Ninou, E., Hwang, E., Cole, M.A., Liu, S.J., Horlbeck, M.A., Chen, J., Replogle, J.M., **Carosso, G.A.**, Eng, N.W.L., Chang, J., Shen, Y., Weissman, J.S., Lim, D.A. (2022). Dual genome-wide coding and lncRNA screens in neural induction of induced pluripotent stem cells. *Cell Genomics*, 2(11), 100177.
3. Zheng, S.C., Stein-O'Brien, G., Augustin, J.J., Slosberg, J., **Carosso, G.A.**, Winer, B., Shin, G., Bjornsson, H.T., Goff, L.A., Hansen, K.D. (2022). Universal prediction of cell-cycle position using transfer learning. *Genome Biology*, 23(1), 41.
4. Breevoort, A., **Carosso, G.A.**, Mostajo-Radji, M.A. (2020). High-altitude populations need special considerations for COVID-19. *Nature Communications*, 11, 3280.

5. **Carosso, G.A.**, Boukas, L., Augustin, J.J., Nguyen, H.N., Winer, B.L., Cannon, G.H., Robertson, J.D., Zhang, L., Hansen, K.D., Goff, L.A., Bjornsson, H.T. (2019). Precocious neuronal differentiation and disrupted oxygen responses in Kabuki syndrome. *JCI Insight*, 4(20), e129375.
6. **Carosso, G.A.**, Ferreira, L.M.R., Mostajo-Radji, M.A. (2019). Scientists as non-state actors of public diplomacy. *Nature Human Behaviour*, 3(11), 1129–1130.
7. **Carosso, G.A.**, Ferreira, L.M.R., Mostajo-Radji, M.A. (2019). Developing brains, developing nations: can scientists be effective non-state diplomats? *Frontiers in Education*, 4, 95.
8. Ferreira, L.M.R., **Carosso, G.A.**, Lopez-Videla, B., Diez, G.V., Rivera-Betancourt, L.I., et al. (2019). Effective participatory science education in a diverse Latin American population. *Palgrave Communications*, 5(1), 11.
9. Benjamin, J.S., Pilarowski, G.O., **Carosso, G.A.**, Zhang, L., Huso, D.L., Goff, L.A., Vernon, H.J., Hansen, K.D., Bjornsson, H.T. (2016). A ketogenic diet rescues hippocampal memory defects in a mouse model of Kabuki syndrome. *PNAS*, 114(1), 125–130.
10. Vanderver, A., Tonduti, D., Bernard, G., Lai, J., Rossi, C., **Carosso, G.**, Quezado, M., Wong, K., Schiffmann, R. (2013). More than hypomyelination in Pol-III disorder. *Journal of Neuropathology & Experimental Neurology*, 72(1), 67–75.
11. Ben-Shahar, O.M., Szumlinski, K.K., Lominac, K.D., Cohen, A., Gordon, E., Ploense, K.L., DeMartini, J., Bernstein, N., Rudy, N.M., Nabhan, A.N., Sacramento, A., Pagano, K., **Carosso, G.A.**, Woodward, N. (2012). Extended access to cocaine self-administration results in reduced glutamate function within the medial prefrontal cortex. *Addiction Biology*, 17(4), 746–757.
12. Bernard, G., Chouery, E., Putorti, M.L., Tétreault, M., Takanohashi, A., **Carosso, G.**, Clément, I., Boespflug-Tanguy, O., Rodriguez, D., Delague, V., et al. (2011). Mutations of POLR3A encoding a catalytic subunit of RNA polymerase Pol III cause a recessive hypomyelinating leukodystrophy. *American Journal of Human Genetics*, 89(3), 415–423.